

chemistry

14 Questions • 3 May 2020 • -1003745219994

1 Q1. The root mean square (R.M.S) velocity of a gas is given by the equation:

A. $C.R.M.S = (E/2M)^{(1/2)}$

B. $C.R.M.S = (2E/M)^{(1/2)}$

C. $C.R.M.S = (3E/2M)^{(1/2)}$

D. $C.R.M.S = (E/3M)^{(1/2)}$

Explanation: $C.R.M.S = (2E/M)^{(1/2)}$ where E is the kinetic energy of the gas molecule and M is the mass of the gas molecule.

2 Q2. The temperature of a gas is given by the equation:

A. $\epsilon \epsilon \epsilon . \text{ } ^\circ \text{C}$

B. $\epsilon \epsilon . \text{ } ^\circ \text{C}$

C. $\epsilon \epsilon . \text{ } ^\circ \text{C}$

D. $\epsilon \epsilon . \text{ } ^\circ \text{C}$

Explanation: The temperature of a gas is given by the equation: $T = \frac{2}{3} \frac{E}{k}$ where T is the temperature, E is the kinetic energy and k is the Boltzmann constant.

3 Q3. The root mean square (R.M.S) velocity of a gas is given by the equation:

A. ϵ

B. ϵ

C. ϵ

D. ϵ

Explanation: The root mean square (R.M.S) velocity of a gas is given by the equation: $C.R.M.S = \sqrt{\frac{2E}{M}}$ where E is the kinetic energy and M is the mass of the gas molecule.

4 Q4. The chemical formula of a gas is given by the equation:

A. H_2SO_4

B. $HClO_4$

C. H_2PO_4

D. HNO_3

Explanation: The chemical formula of a gas is given by the equation: HNO_3 is the chemical formula of nitric acid.

5 Q5. FGD a í 2 3/4 " í ÿ ç ° ® 3/4 § í ¯ ® ç • ë " ÿ ç ! â ° • ° 3/4 ¯ 3/4 ¯ 1/4 " 3/4 ?

- A. SO_x
- B. NO_x
- C. CO,

D. CO

Explanation: FGD $\frac{1}{2} \times 10^6$ lbs/day = 10^6 lbs/day SO_x, NO_x ... $\frac{1}{2} \times 10^6$ lbs/day CO

6 Q6. FGD $\overset{a}{\overset{1}{2}}\overset{3}{4}\overset{5}{6}\overset{7}{8}\overset{9}{10}\overset{11}{12}\overset{13}{14}\overset{15}{16}\overset{17}{18}\overset{19}{20}\overset{21}{22}\overset{23}{24}\overset{25}{26}\overset{27}{28}\overset{29}{30}\overset{31}{32}\overset{33}{34}\overset{35}{36}\overset{37}{38}\overset{39}{40}\overset{41}{42}\overset{43}{44}\overset{45}{46}\overset{47}{48}\overset{49}{50}\overset{51}{52}\overset{53}{54}\overset{55}{56}\overset{57}{58}\overset{59}{60}\overset{61}{62}\overset{63}{64}\overset{65}{66}\overset{67}{68}\overset{69}{70}\overset{71}{72}\overset{73}{74}\overset{75}{76}\overset{77}{78}\overset{79}{80}\overset{81}{82}\overset{83}{84}\overset{85}{86}\overset{87}{88}\overset{89}{90}\overset{91}{92}\overset{93}{94}\overset{95}{96}\overset{97}{98}\overset{99}{100}\overset{101}{102}\overset{103}{104}\overset{105}{106}\overset{107}{108}\overset{109}{110}\overset{111}{112}\overset{113}{114}\overset{115}{116}\overset{117}{118}\overset{119}{120}\overset{121}{122}\overset{123}{124}\overset{125}{126}\overset{127}{128}\overset{129}{130}\overset{131}{132}\overset{133}{134}\overset{135}{136}\overset{137}{138}\overset{139}{140}\overset{141}{142}\overset{143}{144}\overset{145}{146}\overset{147}{148}\overset{149}{150}\overset{151}{152}\overset{153}{154}\overset{155}{156}\overset{157}{158}\overset{159}{160}\overset{161}{162}\overset{163}{164}\overset{165}{166}\overset{167}{168}\overset{169}{170}\overset{171}{172}\overset{173}{174}\overset{175}{176}\overset{177}{178}\overset{179}{180}\overset{181}{182}\overset{183}{184}\overset{185}{186}\overset{187}{188}\overset{189}{190}\overset{191}{192}\overset{193}{194}\overset{195}{196}\overset{197}{198}\overset{199}{200}\overset{201}{202}\overset{203}{204}\overset{205}{206}\overset{207}{208}\overset{209}{210}\overset{211}{212}\overset{213}{214}\overset{215}{216}\overset{217}{218}\overset{219}{220}\overset{221}{222}\overset{223}{224}\overset{225}{226}\overset{227}{228}\overset{229}{230}\overset{231}{232}\overset{233}{234}\overset{235}{236}\overset{237}{238}\overset{239}{240}\overset{241}{242}\overset{243}{244}\overset{245}{246}\overset{247}{248}\overset{249}{250}\overset{251}{252}\overset{253}{254}\overset{255}{256}\overset{257}{258}\overset{259}{260}\overset{261}{262}\overset{263}{264}\overset{265}{266}\overset{267}{268}\overset{269}{270}\overset{271}{272}\overset{273}{274}\overset{275}{276}\overset{277}{278}\overset{279}{280}\overset{281}{282}\overset{283}{284}\overset{285}{286}\overset{287}{288}\overset{289}{290}\overset{291}{292}\overset{293}{294}\overset{295}{296}\overset{297}{298}\overset{299}{300}\overset{301}{302}\overset{303}{304}\overset{305}{306}\overset{307}{308}\overset{309}{310}\overset{311}{312}\overset{313}{314}\overset{315}{316}\overset{317}{318}\overset{319}{320}\overset{321}{322}\overset{323}{324}\overset{325}{326}\overset{327}{328}\overset{329}{330}\overset{331}{332}\overset{333}{334}\overset{335}{336}\overset{337}{338}\overset{339}{340}\overset{341}{342}\overset{343}{344}\overset{345}{346}\overset{347}{348}\overset{349}{350}\overset{351}{352}\overset{353}{354}\overset{355}{356}\overset{357}{358}\overset{359}{360}\overset{361}{362}\overset{363}{364}\overset{365}{366}\overset{367}{368}\overset{369}{370}\overset{371}{372}\overset{373}{374}\overset{375}{376}\overset{377}{378}\overset{379}{380}\overset{381}{382}\overset{383}{384}\overset{385}{386}\overset{387}{388}\overset{389}{390}\overset{391}{392}\overset{393}{394}\overset{395}{396}\overset{397}{398}\overset{399}{399}\overset{400}{400}\overset{401}{401}\overset{402}{402}\overset{403}{403}\overset{404}{404}\overset{405}{405}\overset{406}{406}\overset{407}{407}\overset{408}{408}\overset{409}{409}\overset{410}{410}\overset{411}{411}\overset{412}{412}\overset{413}{413}\overset{414}{414}\overset{415}{415}\overset{416}{416}\overset{417}{417}\overset{418}{418}\overset{419}{419}\overset{420}{420}\overset{421}{421}\overset{422}{422}\overset{423}{423}\overset{424}{424}\overset{425}{425}\overset{426}{426}\overset{427}{427}\overset{428}{428}\overset{429}{429}\overset{430}{430}\overset{431}{431}\overset{432}{432}\overset{433}{433}\overset{434}{434}\overset{435}{435}\overset{436}{436}\overset{437}{437}\overset{438}{438}\overset{439}{439}\overset{440}{440}\overset{441}{441}\overset{442}{442}\overset{443}{443}\overset{444}{444}\overset{445}{445}\overset{446}{446}\overset{447}{447}\overset{448}{448}\overset{449}{449}\overset{450}{450}\overset{451}{451}\overset{452}{452}\overset{453}{453}\overset{454}{454}\overset{455}{455}\overset{456}{456}\overset{457}{457}\overset{458}{458}\overset{459}{459}\overset{460}{460}\overset{461}{461}\overset{462}{462}\overset{463}{463}\overset{464}{464}\overset{465}{465}\overset{466}{466}\overset{467}{467}\overset{468}{468}\overset{469}{469}\overset{470}{470}\overset{471}{471}\overset{472}{472}\overset{473}{473}\overset{474}{474}\overset{475}{475}\overset{476}{476}\overset{477}{477}\overset{478}{478}\overset{479}{479}\overset{480}{480}\overset{481}{481}\overset{482}{482}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- A. i, ii
- B. ii, iii
- C. i, iii

D. i, ii, iii

[illegible]

Q7. $-\frac{3}{4}|\bar{1}^- \quad \bar{1} \bar{A}^{\text{TM}} \bar{1}^- \bar{2} \bar{C}^- \bar{3}/_4 \bar{A} \bar{S} \bar{3}/_4 \bar{A} \bar{O} \bar{1}^- \bar{1}/_4 \bar{1}^- \bar{A} \cdot \bar{1} \bar{1}^2 \bar{C} \textcircled{R} \bar{3}/_4 \bar{A} \cdot \bar{C} \bar{O} \bar{1}^0 \bar{A}^0 \bar{C}^- \text{ (i) } \text{Cr}^3 \text{z} \bar{1}^- \bar{1}/_4 \bar{1}^-$
 $\ll^2 \bar{C} \quad \bar{1} \bar{C} \bar{1} \bar{A}^0 \quad \text{IQ} \bar{1}^- \bar{3}/_4 \bar{1}^- \bar{A} \bar{1} \bar{S} \bar{C} \bar{1}^- \bar{A} \bar{1} \bar{1} \bar{1} \bar{1} \bar{C} \bar{1}^0 \bar{1}^0 \bar{3}/_4, \text{ a } \bar{3}/_4 \bar{1}^- \bar{1}/_4 \text{ (iii) } \text{Cd}^2 \text{z} \bar{1}^- \bar{1}/_4 \bar{1}^- \bar{1}^- \bar{A} \cdot \bar{E} \bar{C}^0 \ll^2 \bar{C} \dots \bar{1} \bar{1} \bar{1} \bar{C} \bar{1}^-$

- A. i, ii
- B. ii, iii
- C. i, iii

D. i, ii, iii

Explanation: , ® , í □ “ ¾ “ Ë ~ ÿ “ ¾ , ç • • ¬ , , í ¬ □ “ í □ í ° á í ° - ¾ ¬ á í ° | ° í ¶ “ • ° Ç d

[illegible]

- A. i, ii
- B. i, iii
- C. ii, iii

D. i, ii, iii

Explanation: • ‡ , ®, í ¢ %^a¾ | ¾" ° a¾" º BOD ¬ Ñ | Í § ¸ • ° ¤ Ç^a¾ ° Ç d

Q9. ETP • Ë" ¬°ÍœÍ⁻ a°¿¶Ë§" •°Ç? (i) • ¿" (ii) π°² (iii) —Í^{-3/4}, À^{-1/4}; "¿šÇ° • Ë"ÿ¿

- A. i, ii
- B. i, iii
- C. ii, iii

D. i, ii, iii

Explanation: $\neg \circ [\text{æ} \bar{i}]^{-\text{a}^\circ} \zeta \P \tilde{\text{E}} \S " \check{\text{C}}^\circ \text{æ} " \bar{i}^- \text{ETP} \cdot \zeta "$, $\square^{\circ 2} \cdot \neg$, $- \bar{i}^{-3/4}, \hat{\text{A}}^{-1/4}$, $\cdot^2 \text{a} \bar{i}^\circ \cdot 3/4^\circ | \hat{\text{A}} \cdot \cdot | \hat{\text{A}}^\circ \cdot \cdot \square \check{\text{C}}^{\text{a} 3/4^\circ} \check{\text{C}} d$

Explanation: $\neq \% \ddot{Y} \acute{O} \ddot{E} \ll \mathcal{C} \cdot \mathfrak{C} \P \text{ " } ^a \frac{3}{4} \text{ " } \mathcal{C} \boxtimes \mathfrak{C} \text{ } ^a \acute{A} \cdot \acute{I} \ddot{Y} \mathcal{C} \cdot ^\circ \text{ } \% ^a \frac{3}{4} | \frac{3}{4} \text{ " } \mathfrak{C} ^\circ \text{ } \textcircled{R} \frac{3}{4} \boxtimes \acute{I} ^\circ \frac{3}{4} \neg \frac{3}{4} ; \frac{1}{4} \frac{3}{4} \overline{\frac{1}{4}} d$

Explanation: $\text{Pb}^{2+} \left(\text{Pb}^{2+} \right) \rightarrow \text{Pb}^{2+} + 2e^-$ $\text{Pb}^{2+} + 2e^- \rightarrow \text{Pb}^{2+}$ $\text{Pb}^{2+} + 2e^- \rightarrow \text{Pb}^{2+}$

Explanation: $\text{NaAlSi}_3\text{O}_8 \cdot 3\text{H}_2\text{O} \rightarrow \text{Na}^+ + \text{Al}^{3+} + 3\text{Si}^{4+} + 8\text{O}^{2-} + 6\text{H}^+ + 3\text{H}_2\text{O}$

Explanation: $(NH_4)_2HPO_4$ $1 \frac{1}{2} \times \frac{3}{4} = \frac{9}{8} \times \frac{3}{4} = \frac{27}{32} \times \frac{3}{4} = \frac{81}{128} \times \frac{3}{4} = \frac{243}{512} \times \frac{3}{4} = \frac{729}{2048} \times \frac{3}{4} = \frac{2187}{8192} \times \frac{3}{4} = \frac{6561}{32768} \times \frac{3}{4} = \frac{19683}{131072} \times \frac{3}{4} = \frac{59049}{524288} \times \frac{3}{4} = \frac{177147}{2097152} \times \frac{3}{4} = \frac{531441}{8388608} \times \frac{3}{4} = \frac{1594323}{33554432} \times \frac{3}{4} = \frac{4782969}{134217728} \times \frac{3}{4} = \frac{14348907}{536870912} \times \frac{3}{4} = \frac{43046721}{2147483648} \times \frac{3}{4} = \frac{129140163}{8589934592} \times \frac{3}{4} = \frac{387420489}{34359738368} \times \frac{3}{4} = \frac{1162261467}{137438953472} \times \frac{3}{4} = \frac{3486784401}{549755813888} \times \frac{3}{4} = \frac{10460353203}{2199023255552} \times \frac{3}{4} = \frac{31381059609}{8796093022208} \times \frac{3}{4} = \frac{94143178827}{35184372088832} \times \frac{3}{4} = \frac{282429536481}{140737488355328} \times \frac{3}{4} = \frac{847288609443}{562949953421248} \times \frac{3}{4} = \frac{2541865828329}{2251799813684992} \times \frac{3}{4} = \frac{7625597484987}{9007199254739968} \times \frac{3}{4} = \frac{22876792454961}{36028797018959872} \times \frac{3}{4} = \frac{68630377364883}{144115188075839488} \times \frac{3}{4} = \frac{205891132094649}{576460752303357952} \times \frac{3}{4} = \frac{617673396283947}{2305843009213431808} \times \frac{3}{4} = \frac{1853020188851841}{9223372036853727232} \times \frac{3}{4} = \frac{5559060566555523}{36893488151414908928} \times \frac{3}{4} = \frac{16677181699666569}{147573952605659635712} \times \frac{3}{4} = \frac{50031545098999707}{590295810422638542848} \times \frac{3}{4} = \frac{150094635296999121}{2361183241690554171392} \times \frac{3}{4} = \frac{450283905890997363}{9444732966762216685568} \times \frac{3}{4} = \frac{1350851717672992089}{37778931867048866742272} \times \frac{3}{4} = \frac{4052555153018976267}{151115727468195466969088} \times \frac{3}{4} = \frac{12157665459056928801}{604462909872781867876352} \times \frac{3}{4} = \frac{36472996377170786403}{2417851639491127471505408} \times \frac{3}{4} = \frac{109418989131512359209}{9671406557964509886021632} \times \frac{3}{4} = \frac{328256967394537077627}{38685626231858039544086528} \times \frac{3}{4} = \frac{984770902183611232881}{154742504927432158176346112} \times \frac{3}{4} = \frac{2954312706550833698643}{618970019709728632705384448} \times \frac{3}{4} = \frac{8862938119652501095929}{2475880078838914530821537792} \times \frac{3}{4} = \frac{26588814358957503287787}{9903520315355658123286151168} \times \frac{3}{4} = \frac{79766443076872509863361}{39614081261422632493144604672} \times \frac{3}{4} = \frac{239299329230617529580083}{158456325045690529972578418688} \times \frac{3}{4} = \frac{717897987691852588740249}{633825300182762119890313674752} \times \frac{3}{4} = \frac{2153693963075557766220747}{2535301200731048479561254698944} \times \frac{3}{4} = \frac{6461081889226673308662241}{10141204802924193918245018795776} \times \frac{3}{4} = \frac{19383245667680019925986723}{40564819211696775672980075183104} \times \frac{3}{4} = \frac{58149737003040059777960169}{162259276846787102691920300732416} \times \frac{3}{4} = \frac{174449211009120179333880507}{649037107387148410767681202929664} \times \frac{3}{4} = \frac{523347633027360537901641521}{2596148429548593643070724811718656} \times \frac{3}{4} = \frac{1570042899082081613704924563}{10384593718194374572282899246874624} \times \frac{3}{4} = \frac{4710128697246244841114773689}{4153837487277749828913159698749856} \times \frac{3}{4} = \frac{14130386091738734523344321067}{16615349949110999315652638794999424} \times \frac{3}{4} = \frac{42391158275216203570032963201}{66461399796443997262610555179997696} \times \frac{3}{4} = \frac{127173474825648610710098889603}{265845599185775989050442220719990784} \times \frac{3}{4} = \frac{381520424476945832130296668809}{1063382396743103956201768882879963136} \times \frac{3}{4} = \frac{1144561273430837496390889906427}{4253529586972415824807075531519852544} \times \frac{3}{4} = \frac{3433683820292512489172669719281}{17014118347889663299228302126079410176} \times \frac{3}{4} = \frac{10301051460877537467518009157843}{68056473391558653196913208504317640704} \times \frac{3}{4} = \frac{30903154382632612402554027473529}{272225893566234612787652834017270562816} \times \frac{3}{4} = \frac{92709463147897837207662082420587}{1088903574264938451150611336069082251264} \times \frac{3}{4} = \frac{278128389443693511622986247261761}{4355614297059753804602445344276329005056} \times \frac{3}{4} = \frac{834385168330980534868958741785283}{17422457188239015218409781377105316020224} \times \frac{3}{4} = \frac{2503155504992941604606876225355849}{69689828752956060873639125508421264080896} \times 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Explanation: $\cdot^{-}\frac{1}{4}{}^2\frac{3}{4}\square\zeta$ As $\frac{3}{4}\S\frac{3}{4}{}^\circ\pounds\square t{}^\circ\acute{\imath}\zeta{}^{\circ}\ddot{\text{E}}{}^{\text{a}}\frac{3}{4}\neq{}^\circ\frac{3}{4}\neq\ddot{\text{Y}}t\cdot\frac{3}{4}{}^\circ\zeta\neq\frac{3}{4}\cdot\zeta d$